

14 Singular Value Decomposition



14.1 Singular Value Decomposition

Any real matrix of any shape factors into rotate, scale, rotate again — revealing the matrix's inner structure

Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master in This Section

1

A real matrix of any shape admits a singular value decomposition (SVD)

2

In full SVD, both U and V are orthogonal matrices

3

U and V each form an orthonormal basis for their own space — think of them as rotations

4

S is a diagonal matrix; its main diagonal holds the singular values

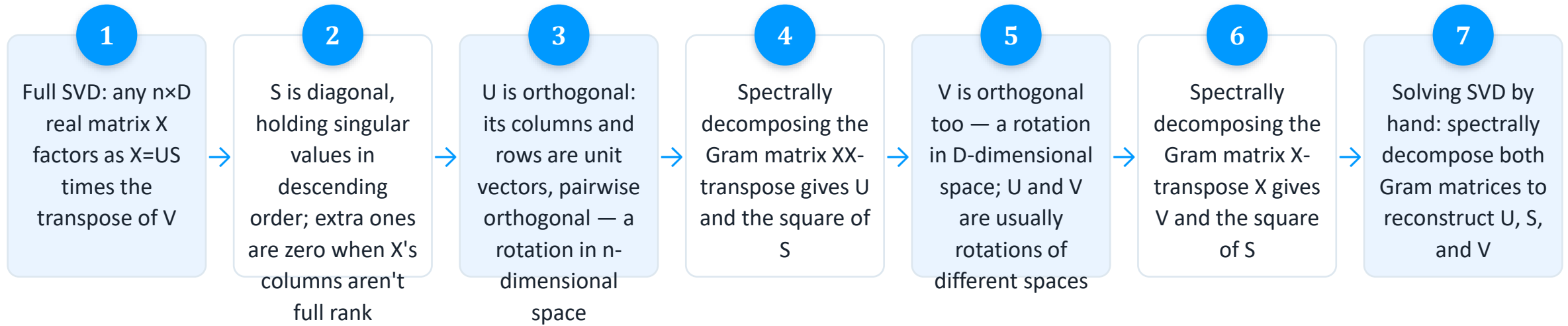
5

The rank of a matrix equals its number of nonzero singular values

6

Solving the eigenvalues and eigenvectors of two Gram matrices reconstructs the SVD

Rotate → Scale → Rotate Again



Understanding Singular Value Decomposition from Several Angles

① Algebraic lens: $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$ — any real matrix of any shape admits an SVD

② Orthogonal lens: both $\mathbf{U}$ and $\mathbf{V}$ are orthogonal matrices, viewable as space rotations

③ Scaling lens: $\mathbf{S}$ is diagonal, and its singular values represent the scaling effect

④ Rank lens: a matrix's rank equals its number of nonzero singular values

⑤ Decomposition lens: spectrally decomposing $\mathbf{X}\mathbf{X}^T$ gives $\mathbf{U}$; $\mathbf{X}^T\mathbf{X}$ gives $\mathbf{V}$

⑥ Computation lens: hand-solving SVD hinges on spectrally decomposing two Gram matrices

Given Any Real Matrix X , Full Singular Value Decomposition Gives

$$X_{n \times D} = U_{n \times n} S_{n \times D} V^T_{D \times D}$$

- U is the left-singular-vector matrix: orthogonal, columns spanning an orthonormal basis for $\mathbb{R}^n$
- S is diagonal, holding the singular values; V is the right-singular-vector matrix, also orthogonal

Figure 1 · A Heatmap of Full SVD

- $X=U@S@V$ -transpose: the colored cells make the numeric values of each factor visible at a glance

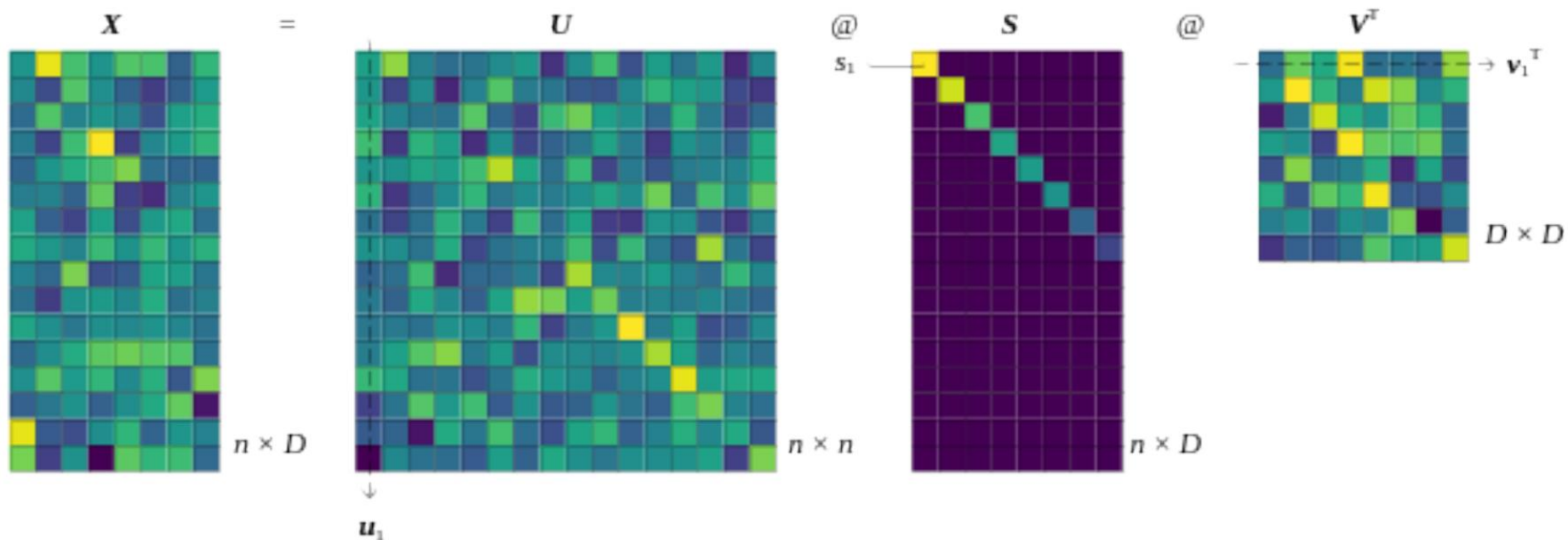
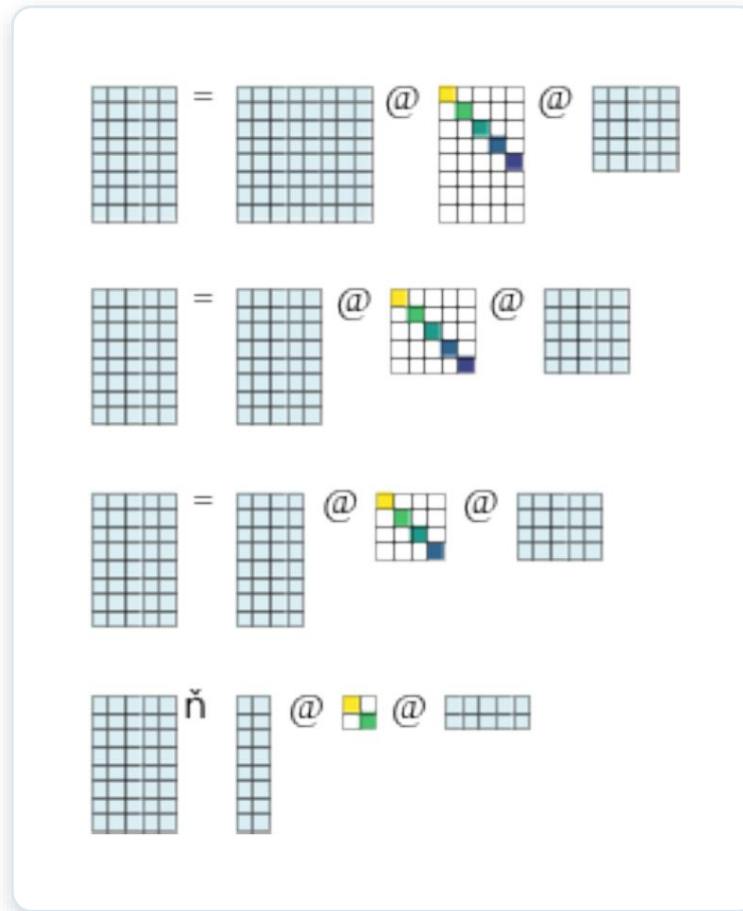


图 1. 完全型 SVD 分解

Figure 2 · Four Types of SVD

- Beyond full SVD there are economy, compact, and truncated SVD — covered later in this chapter



Singular Values Are Usually Listed in Descending Order

$$s_1 \geq s_2 \geq \dots \geq s_D > 0$$

- If X has full column rank, all singular values are positive; with rank r , $D-r$ of them are zero
- In other words, the rank of a matrix equals its number of nonzero singular values

Figure 3 · Full SVD When X Doesn't Have Full Column Rank

- Replacing X's last two columns with its first two zeroes out two singular values

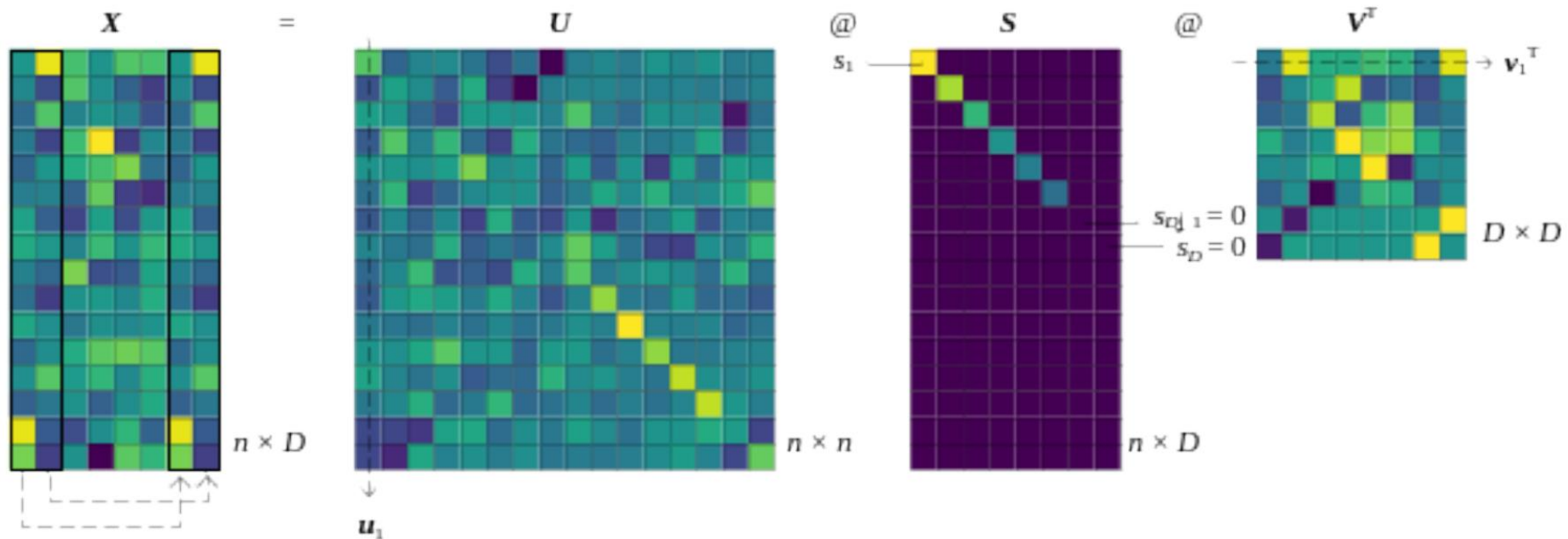


图 3. 完全型 SVD 分解，X 列不满秩

In Full SVD, the Left-Singular-Vector Matrix U Is Orthogonal

$$U^T @ U = U @ U^T = I$$

- Both U's columns and its rows are unit vectors, pairwise orthogonal
- Geometrically, orthogonal matrix U can be viewed as a rotation in n-dimensional space

Figure 4 · U's Columns Are Unit Vectors, Pairwise Orthogonal

- The transpose of U times U equals the identity I — the diagonal is all 1s, everything else 0

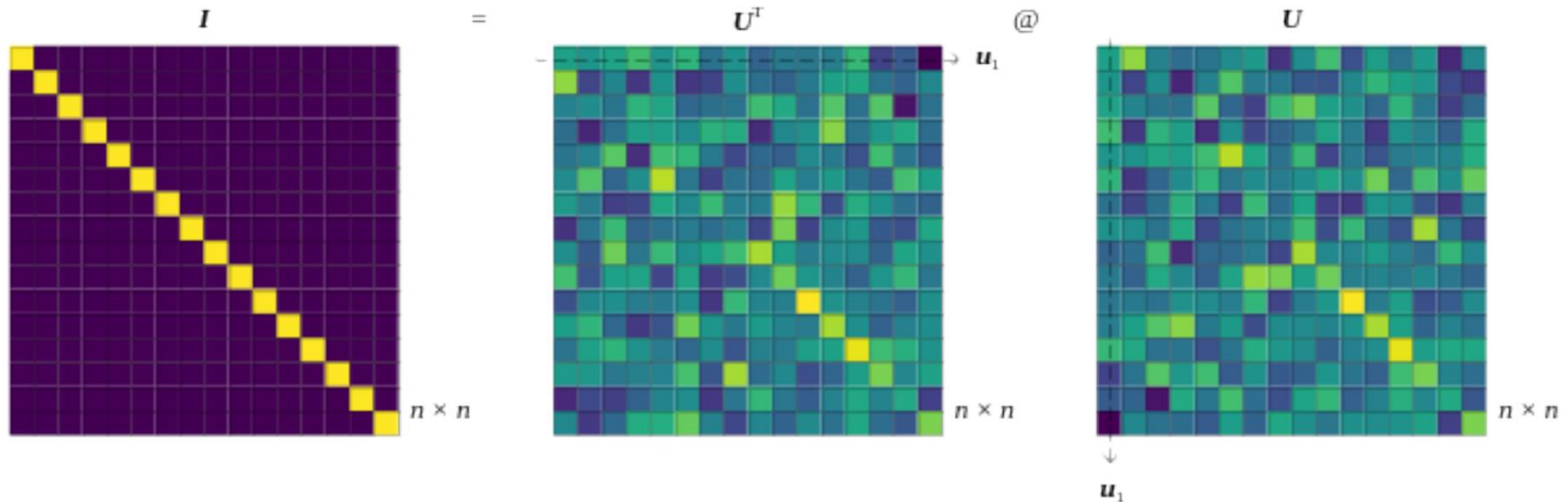


图 4. U 列向量均为单位向量，且两两正交

Figure 5 · U's Rows Are Unit Vectors, Pairwise Orthogonal

- U times the transpose of U also equals the identity I , so the rows are pairwise orthogonal too

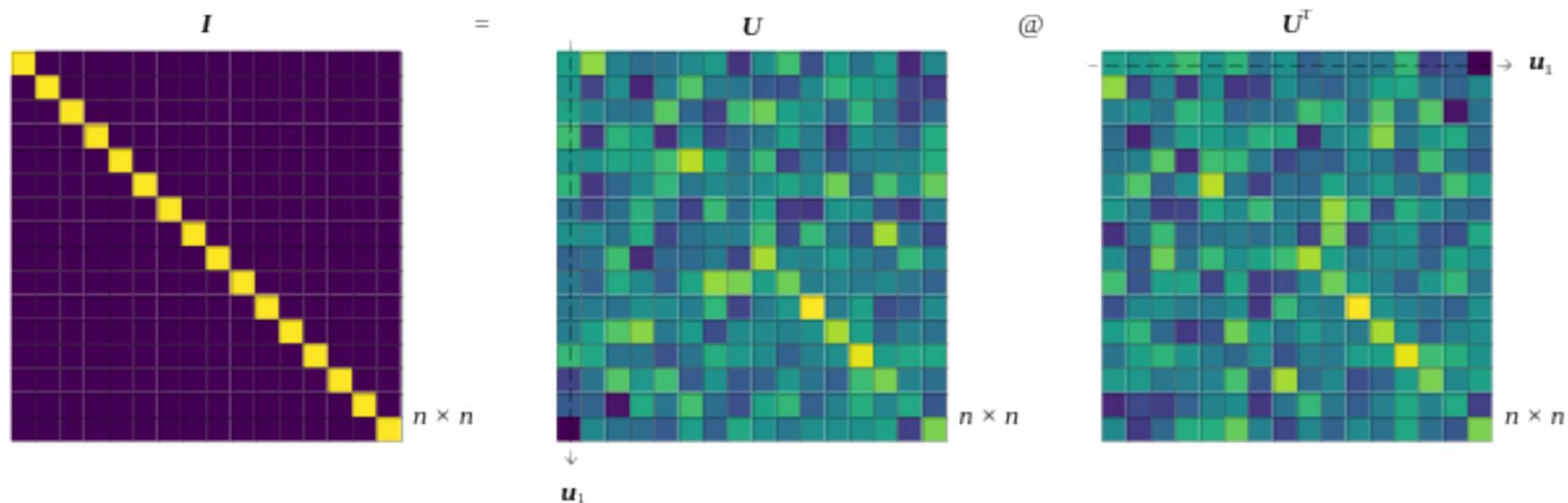


图 5. U 行向量均为单位向量，且两两正交

XX-Transpose Is the Gram Matrix of X-Transpose

$$XX^T = USS^TU^T$$

- A Gram matrix is symmetric and positive semi-definite (or definite), so this is a spectral decomposition
- Expanding it out, a great many of XX-transpose's eigenvalues turn out to be zero

Figure 6 · Computing the Gram Matrix XX -Transpose

- XX -transpose = X @ the transpose of X , an $n \times n$ symmetric matrix

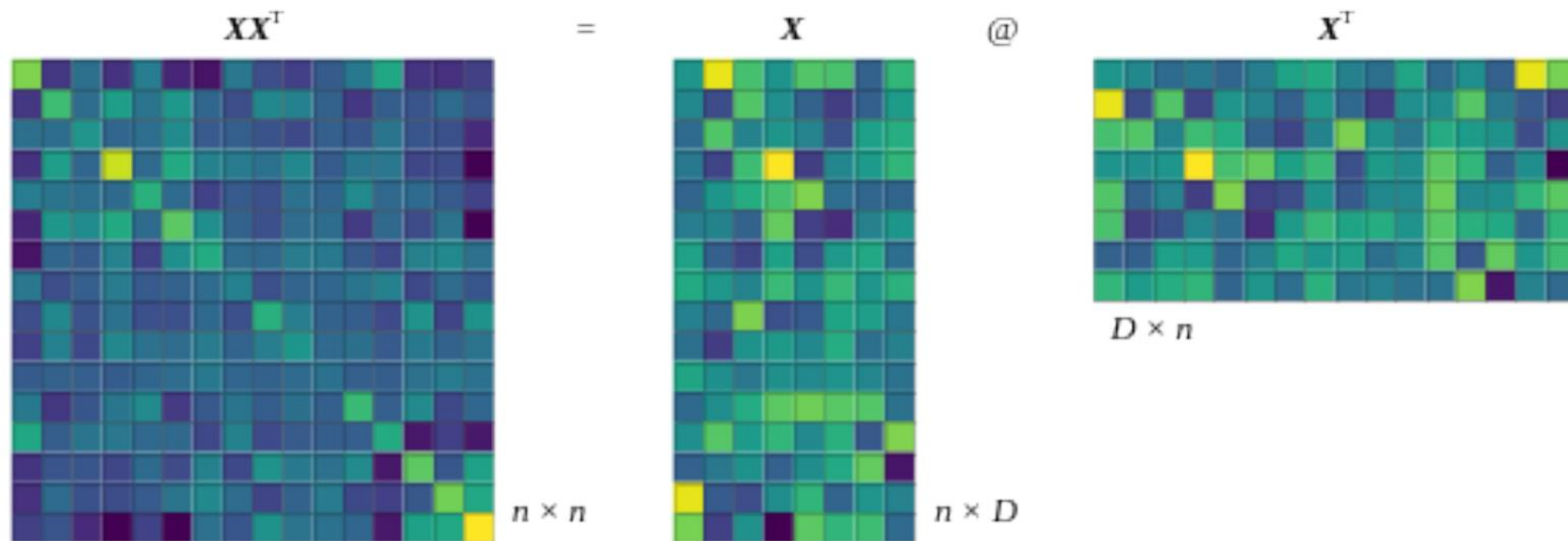


图 6. 计算格拉姆矩阵 XX^T

Figure 7 · Eigen Decomposition of XX^T -Transpose, Full Column Rank

- With X's singular values all nonzero, XX -transpose ends up with $n-D$ zero eigenvalues

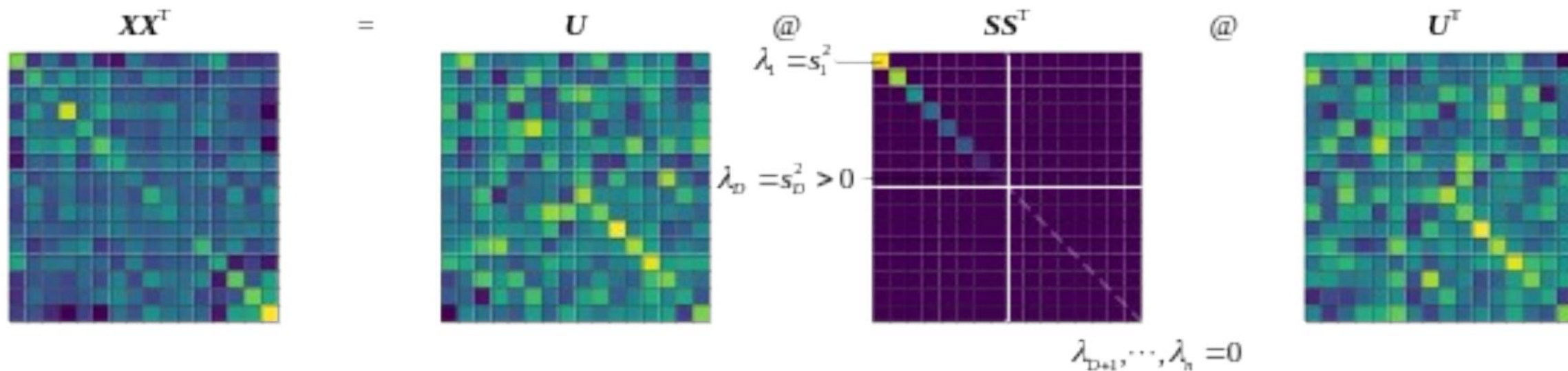


图 7. XX^T 的特征值分解, X 列满秩

Figure 8 · Eigen Decomposition of XX^T -Transpose, Rank-Deficient

- With X at rank r , XX -transpose ends up with $n-r$ zero eigenvalues

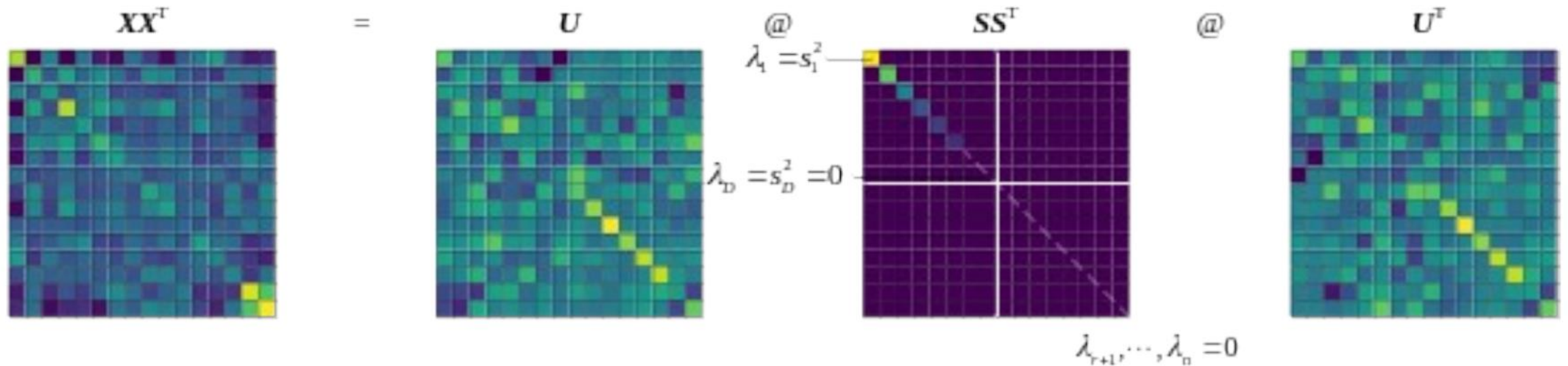


图 8. XX^T 的特征值分解, X 列不满秩

In Full SVD, the Right-Singular-Vector Matrix V Is Orthogonal

$$V^T @ V = V @ V^T = I$$

- Geometrically, orthogonal matrix V can be viewed as a rotation in D -dimensional space
- Note that U and V are, in general, rotations of different spaces

Figure 9 · Right-Singular-Vector Matrix V Is Orthogonal

- The transpose of V times V , and V times the transpose of V , both equal the identity I

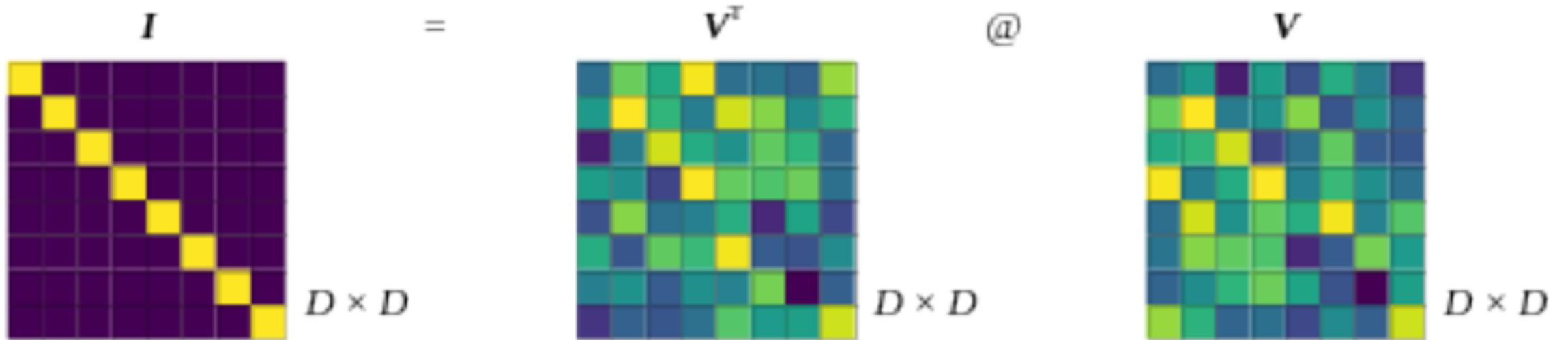


图 9. 右奇异向量矩阵 V 为正交矩阵

X -Transpose X Is the Gram Matrix of X

$$X^T X = V S^T S V^T$$

- This is also a spectral decomposition — this book covered it earlier when introducing Gram matrices
- Comparing the two Gram matrices, they share the same nonzero eigenvalues

Figure 10 · Computing the Gram Matrix X-Transpose X

- X-transpose X = the transpose of X @ X, a $D \times D$ symmetric matrix

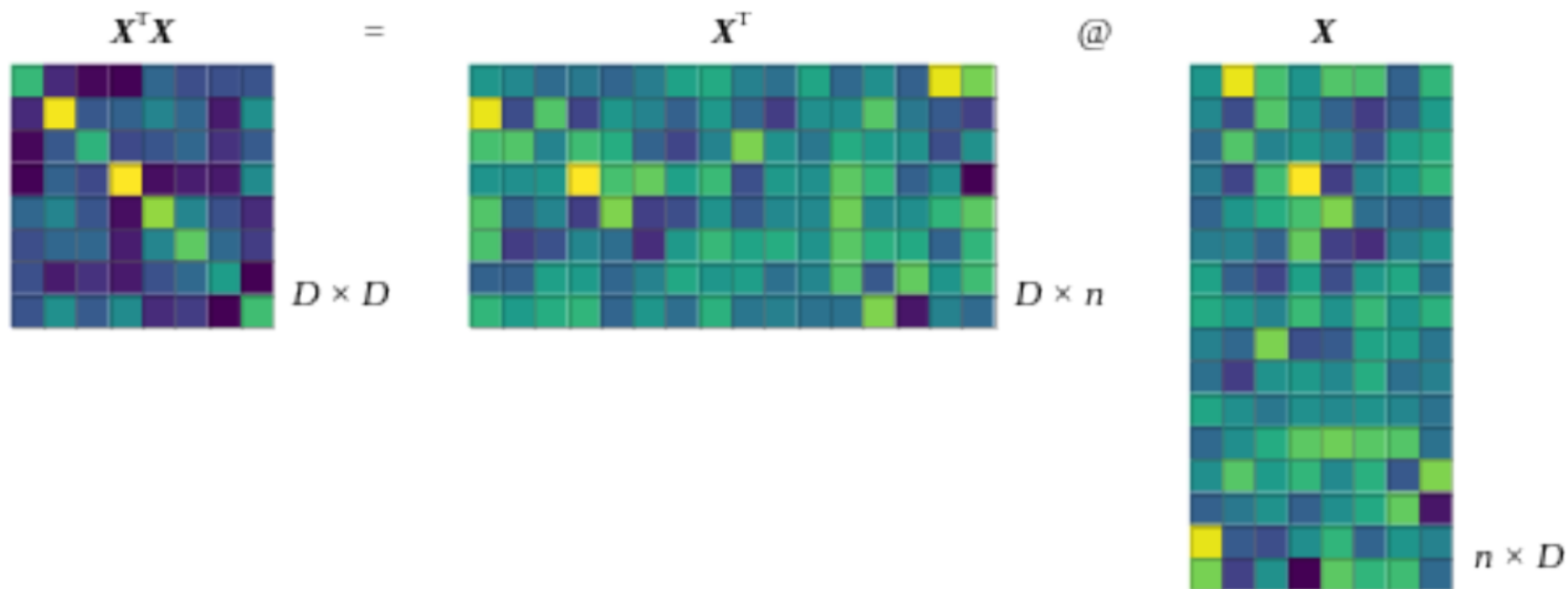


图 10. 计算格拉姆矩阵 $X^T X$

Figure 11 · Eigen Decomposition of X-Transpose X, Full Column Rank

- With X at full column rank, all of X-transpose X's eigenvalues are nonzero



图 11. $X^T X$ 的特征值分解, X 列满秩

Figure 12 · Eigen Decomposition of X-Transpose X, Rank-Deficient

- With X at rank r, X-transpose X ends up with D-r zero eigenvalues



图 12. $X^T X$ 的特征值分解, X 列不满秩

Given a Tall, Narrow Matrix A, Compute the First Gram Matrix

$$A = [0,1; 1,1; 1,0] , \quad A^T A = [2,1; 1,2]$$

- Figures 7 and 11 point the way to hand-solving SVD — spectrally decompose two Gram matrices

Eigenvalues and Eigenvectors of the Transpose of A Times A

$$\lambda_1 = 3, \quad \mathbf{v}_1 = [\sqrt{2}/2; \sqrt{2}/2], \quad \lambda_2 = 1, \quad \mathbf{v}_2 = [-\sqrt{2}/2; \sqrt{2}/2]$$

- For a refresher on hand-solving eigen decomposition, see Section 11.1 of this book

Compute the Second Gram Matrix, A Times the Transpose of A

$$AA^T = [1,1,0; 1,2,1; 0,1,1]$$

- Note the shapes differ: the transpose of A times A is 2×2, while A times its transpose is 3×3

Eigenvalues and Eigenvectors of A Times the Transpose of A (Part 1)

$$\lambda_1 = 3, \quad \mathbf{u}_1 = [1/\sqrt{6}; 2/\sqrt{6}; 1/\sqrt{6}]$$

- λ_1 and λ_2 show up in both Gram matrices — this is no coincidence

Eigenvalues and Eigenvectors of A Times the Transpose of A (Part 2)

$$\lambda_2 = 1, \quad \mathbf{u}_2 = [\sqrt{2}/2; 0; -\sqrt{2}/2], \quad \lambda_3 = 0, \quad \mathbf{u}_3 = [\sqrt{3}/3; -\sqrt{3}/3; \sqrt{3}/3]$$

- $\lambda_3=0$ and its $\mathbf{u}_3$ are exactly the extra eigenpair that only shows up in A times its transpose

The Diagonal Entries of Singular Value Matrix S Are Square Roots of the Eigenvalues

$$S = [\sqrt{\lambda_1}, 0; 0, \sqrt{\lambda_2}; 0, 0] = [\sqrt{3}, 0; 0, 1; 0, 0]$$

- S's diagonal entries are the square roots of λ_1 and λ_2 — a key distinction, and connection, between eigen decomposition and SVD

The Full Singular Value Decomposition of Matrix A

$$A = USV^T = \begin{bmatrix} 1/\sqrt{6}, \sqrt{2}/2, \sqrt{3}/3; 2/\sqrt{6}, 0, -\sqrt{3}/3; 1/\sqrt{6}, -\sqrt{2}/2, \sqrt{3}/3 \end{bmatrix} \begin{bmatrix} \sqrt{3}, 0; 0, 1; 0, 0 \end{bmatrix} \begin{bmatrix} \sqrt{2}/2, \sqrt{2}/2; -\sqrt{2}/2, \sqrt{2}/2 \end{bmatrix}^T$$

- Hand-solving SVD isn't the goal — what matters is grasping the ideas and viewing angles behind it

What Else Singular Value Decomposition Can Do

- ① Data compression: keeping just the largest few singular values approximately reconstructs a matrix
- ② Feature extraction: the size of a singular value reflects how important its direction is
- ③ Recommender systems: matrix factorization techniques are often built on SVD
- ④ Image processing: low-rank approximation is used for image compression and denoising
- ⑤ Natural language processing: latent semantic analysis (LSA) uses SVD to uncover latent structure between words and documents
- ⑥ Numerical computing: computing a matrix's pseudoinverse or condition number both rely on SVD

Table · XX-Transpose vs. X-Transpose X

Dimension	Gram Matrix XX-Transpose	Gram Matrix X-Transpose X
Matrix shape	$n \times n$	$D \times D$
Corresponding singular vectors	U (left-singular vectors)	V (right-singular vectors)
Eigenvalue-to-singular-value relation	eigenvalue = singular value squared	eigenvalue = singular value squared
Nonzero eigenvalues (X full column rank)	D	D (all nonzero)
Zero eigenvalues (X full column rank)	$n - D$	0

SUMMARY

Key Takeaways on Full SVD

Any real matrix X of any shape factors as $X=USV^T$

U and V are both orthogonal — viewable as rotations; S is diagonal, holding the singular values

A matrix's rank equals its number of nonzero singular values

Spectrally decomposing XX -transpose gives U and S^2 ; X -transpose X gives V and S^2

Hand-solving SVD hinges on spectrally decomposing two Gram matrices to reconstruct U , S , V

Key Terms in This Section

Singular Value Decomposition *SVD* — $X=USV^T$, a universal decomposition for any real matrix

Left-Singular-Vector Matrix U — orthogonal, columns spanning an orthonormal basis for $\mathbb{R}^n$

Gram Matrix XX^T or X^TX — symmetric and positive semi-definite or definite

Rank Equals a matrix's number of nonzero singular values

Full SVD The SVD form where U and V are both fully orthogonal square matrices

Singular Value S 's diagonal entries — the scaling effect along each dimension

Right-Singular-Vector Matrix V — orthogonal, columns spanning an orthonormal basis for $\mathbb{R}^D$

Spectral Decomposition The eigen decomposition of a real symmetric matrix

Orthogonal Matrix Its transpose times itself equals the identity — geometrically, a rotation

Economy SVD Coming up next — U becomes a semi-orthogonal matrix

FURTHER LEARNING

Suggested Reading

Coming Up · 14.2 A Geometric View of SVD

Chapter 14 of this book — understanding U , S , V as geometric transformations

Review · 11.1 Eigen Decomposition

Chapter 11 of this book — how to hand-solve eigen decomposition

Review · Gram Matrices and Spectral Decomposition

Earlier in this book — the definition, properties, and spectral decomposition of Gram matrices

Open-Source Repository

Code and figures for the full book: <https://github.com/Visualize-ML>

Exercises Q1 – Q9

Q1 By hand, find the SVD of $A=[0,-1;-1,-1;-1,0]$, and verify with Python.

Q3 By hand, find the SVD of $A=[0,1,1;1,0,1;1,1,0]$, and verify with Python.

Q5 Use `numpy.linalg.svd()` to decompose this section's tall matrix $A=[0,1;1,1;1,0]$, and compare with the hand calculation.

Q7 Prove that the singular values of any real matrix X are non-negative real numbers.

Q9 Explore: if matrix X is square and symmetric, how does SVD relate to spectral decomposition?

Q2 By hand, find the SVD of $A=[5/4,-3/4;-3/4,5/4]$, and verify with Python.

Q4 By hand, find the SVD of $A=[0,1,1;1,1,0]$, and verify with Python.

Q6 Verify whether this section's singular values $s_1=\sqrt{3}$ and $s_2=1$ equal the square roots of the eigenvalues of the transpose of A times A .

Q8 Explain why XX -transpose and X -transpose X share the same nonzero eigenvalues.

Exercises Q10 – Q17

Q10 Use Python to verify that orthogonal matrix U satisfies: the transpose of U times U equals the identity.

Q12 Prove that a matrix X 's rank equals its number of nonzero singular values.

Q14 Write a Python function that implements hand-solved SVD: spectrally decompose two Gram matrices to reconstruct U , S , V .

Q16 Compute the SVD of a 4×3 matrix, and verify that both U and V are orthogonal.

Q11 Explore: why can U and V in full SVD be viewed as space rotations? Discuss in terms of their determinants.

Q13 Explore: Figure 3 shows two zero singular values when X 's columns aren't full rank — what geometric property of X does this correspond to?

Q15 Explore: how do SVD and eigen decomposition differ? Which matrices admit both?

Q17 Explore: how does the magnitude of a singular value relate to a matrix's "important directions"?

Exercises Q18 – Q25

Q18 Use matplotlib to visualize how this section's matrix A transforms a unit circle, marking the singular-value directions.

Q20 Explore: how does SVD connect to principal component analysis (PCA)?

Q22 Explore: why does this section say "any real matrix of any shape admits an SVD," while eigen decomposition only applies to square matrices?

Q24 Explore: how does SVD's range of applications differ from this book's earlier Cholesky and LDL decompositions?

Q19 Explore: how is SVD applied to data compression? Research and summarize the underlying idea.

Q21 Prove that if X is an orthogonal matrix, all of its singular values equal 1.

Q23 Use Python to compare, on the same matrix, eigen decomposition (where it exists) versus SVD — where do the results differ?

Q25 Preview question: the next section takes a geometric view of SVD — what geometric operation do you think each of U , S , V corresponds to?

10 Prompts to Explore with AI

1 Explain why a real matrix of any shape admits an SVD, while eigen decomposition only applies to square matrices.

2 Explain what U , S , and V each mean in full SVD, and their respective matrix shapes.

3 Explain why an orthogonal matrix can be viewed as a rotation in space.

4 Explain why a matrix's rank equals its number of nonzero singular values.

5 Use Python to demonstrate how to compute a matrix's SVD with `numpy.linalg.svd()`.

10 Prompts to Explore with AI

6 Explain the respective roles that the Gram matrices XX^T and $X^T X$ play in SVD.

7 Explain why the diagonal entries of singular value matrix S are square roots of the Gram matrix's eigenvalues.

8 Explain the full process of hand-solving SVD, from the Gram matrices to U , S , V .

9 Explain the connection and the difference between SVD and this book's earlier spectral decomposition.

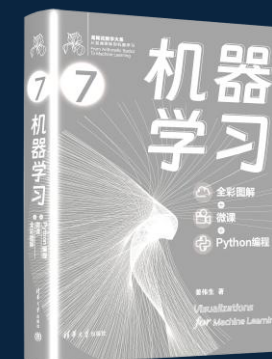
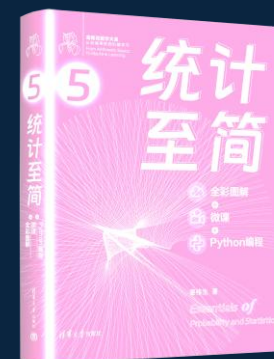
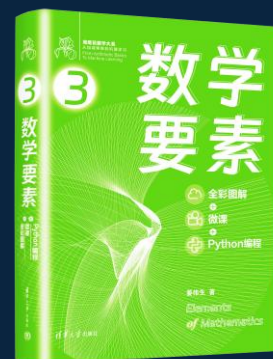
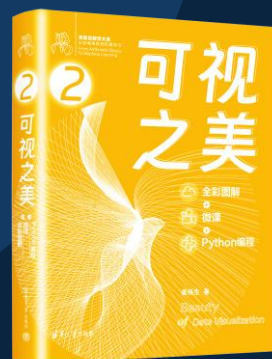
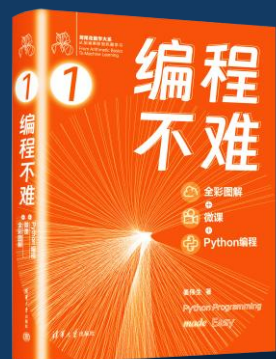
10 Explain possible applications of SVD in data compression or feature extraction.

数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析
图和网络

回归、降维
分类、聚类

实践

